

Measuring solar cycles and the impact of solar winds on galactic cosmic rays

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1 Aim

The aim of this experiment is to determine, with more detail, how the sun's solar cycle behaves and what the impacts of the sun's shifting magnetic field has upon galactic cosmic rays entering the solar system.

2 Introduction

The sun is one of the most studied objects within the history of humankind. From its inclusion into religious figures, such as Ra in Egyptian mythology, to philosophical interpretations of its function and purpose. In this experiment, the focus is placed on the cycles the sun passes through, the effects the cycle has on the heliosphere and any cosmic rays that enter or originate within our solar system.

The sun constantly emits charged particles from itself. These collections of charged particles are called "solar wind"[5]. These solar winds collide with other mediums (such as planets) but sometimes reach beyond the orbit of Pluto. This creates a massive magnetic field around the solar system that helps shield the solar system from galactic cosmic rays that may intercept with the bodies within the solar system. This then provides the earth with an even stronger layer of protection, in addition to its own magnetic field that protects it from solar flairs and other cosmic threats.

This heliosphere also effects any cosmic rays that may have entered or originated within our solar system. Cosmic rays are charged particles (such as protons, electrons, positrons, etc) that travel at high speeds and originate from many origins[2]. Some come from our sun and form part of the solar winds that shield the solar system. Another source of cosmic rays come from the local planet Jupiter, whose magnetosphere give off high energy protons (known as Jovian cosmic ray protons (JCRP))[7].

Finally, the most energetic cosmic rays to have entered the solar system are galactic cosmic rays (GCR). These cosmic rays are typically created by events that release incredibly large amounts of energy (events such as supernovas, which is the event where a star explodes) and are subsequently emitted from the event. Often times when they do arrive to the solar system, they are fully stripped of their electrons, making them ionized [6]. Because they then only contain the positively charged protons within their nucleus (alongside neutrons, which are not charged particles), these GCR's are very susceptible to a magnetic field, or in other words, are modulated easily by magnetic forces that are present within the solar system.

Typically, when a GCR enters Earth atmosphere, it does it directly impact the crust of the earth or any objects on surface. Rather, the GCR collides with particles in the atmosphere (such as nitrogen, oxygen, etc). These atoms then become highly energised and in turn collide with other atmospheric atoms,

leading to a cascading event of energised atoms that can then be detected on the surface of the earth [6].

One important subject regarding the sun is the solar cycle the sun undergoes every 11 years. This cycle dictates the amount of activity the sun undergoes in the current time period. For example, if the sun is in a higher activity part of the solar cycle, then it will emit more ultraviolet and X-Ray waves, which pose a threat to satellites currently orbiting earth [3]. Within the sun is also a constant back and forth between the plasma that moves around the on the surface and the magnetic field generated by the sun. The plasma in the interior of the sun moves around, and because of the charged particles within the plasma, generate its own magnetic field as well that typically counter-act the magnetic field of the sun [3]. Thus, increased plasma activity on the sun will have strong effects of the magnetic field of the sun.

As a result of the movement of plasma across the sun, sunspots can form on the face of the sun during the 11 year solar cycle. These sunspots arise due to how the movement of the plasma "twists" the magnetic field lines at certain spots. Due to this magnetic field that has been shaped by the plasma, it can often inhibit the heat within the deeper interior from rising to that point on the surface, leading to a cooler region [4]. Because of this phenomena, higher plasma activity on the sun can lead to an increase in the strength of the magnetic field, and often gives rise to more sun spots. Areas of stronger magnetic field can also give rise to more solar flares and coronal mass ejections (CME), both of which contribute to the amount of solar wind ejected by the sun [1].

As one can see, the activity level of the sun (dictated by the period within the solar cycle) can modulate the amount of solar wind emissions by the sun, and that in turn effects the modulation of any GCRs that may enter solar system subspace. During higher activity levels in the solar cycle, stronger magnetic fields are generated (which can be observed by measuring the amount of sunspots that appear across the surface), which should then lead to more solar wind being ejected.

Measurements used within this report originate from two sources. The first of which is a yearly averaged cosmic ray proton intensity values, taken by IMP-8 at a rigidity value of $P = 1.28$ GV, and magnetic field values of the sun are used from the OMNI-dataset.

3 Analyse

This analyses shall be separated into two subsections: Mathematical derivations and applications/techniques used to interpolate data.

3.1 Mathematical derivations

To start, the Parker-Transport equation which models a cosmic ray, is denoted by:

$$\frac{\partial f_0}{\partial t} = \nabla \cdot \left(\vec{K} \cdot \nabla f_0 \right) - \vec{v}_{sw} \cdot \nabla f_0 + \frac{1}{3} \left(\nabla \cdot \vec{v}_{sw} \right) \frac{\partial f_0}{\partial \ln \rho} + Q \quad (1)$$

This equation will be operated on using the following assumptions:

- The cosmic ray is in a steady state, so $\frac{\partial f_0}{\partial t} = 0$.
- Adiabatic energy losses are ignored, so $\frac{1}{3} (\nabla \cdot \vec{v}_{sw}) \frac{\partial f_0}{\partial \ln \rho} = 0$
- Assume that the diffusion coefficient κ is only dependent on position r and momentum.
- Assume that the whole system is spherically symmetrical.

Applying these assumptions reduces the Parker-Transport equation to the following expression:

$$v_{sw} f_0 = \kappa \frac{\partial f_0}{\partial r} \quad (2)$$

Solving this first-order differential equation, and solving it in terms of the variable j yields

$$j = j b e^{-M} \quad (3)$$

With M as the modulation parameter, which describes the extent to which a cosmic ray is effected by external factors. Modulation parameter can also be expressed as:

$$M = \int_{r_o}^{r_b} \frac{v_{sw}}{\kappa} dr \quad (4)$$

When the results were processed, the modulation parameter, M , was calculated by rearranging the differential equation solution in terms of M , yielding

$$M = -\ln \left(\frac{j}{j b} \right) \quad (5)$$

Now for the variable κ , this can be expressed as

$$\kappa = a \left(\frac{\bar{B}}{B} \right)^\gamma \quad (6)$$

Later on, calculations for the value γ will be performed, in which case we substitute the κ equation into the integral modulation parameter equation. After

performing the integral (with the assumption that κ and v_{sw} are independent of position, γ solves to

$$\gamma = \log_{\frac{B}{B_0}} \left(84 \frac{v_{sw}}{aM} \right) \quad (7)$$

Applying these values with the given dataset will yield all the needed values.

3.2 Applications used for interpolating data

The dataset given consists of 3 columns of values:

- Year
- Cosmic ray energy intensity of protons (in units $\text{m}^{-2}\text{s}^{-1}\text{sr}^{-1}\text{MeV}^{-1}$, or j in this context)
- Magnetic field strength, in nano Teslas (nT)

Which has been given on a text file. Excel was then used to store the data and further extract information from it.

For graphing of results to show comparisons in values, python was used. More specifically, the following libraries were used to both extract data from the excel application, and also plot the data visually:

- seaborn
- matplotlib
- pandas

With matplotlib and seaborn both acting as libraries specifically for plotting data, and pandas acting as a library that extracts data from various file types (in this case, excel) and also makes it much easier to extract and manipulate data for whatever purposes required.

Seaborn and matplotlib both function very similarly, but matplotlib was used over seaborn later during data interpolation as seaborn took a very long time to process and print graphs.

The libraries were applied as follows

1. Download all the needed data and store it next to where the python files will be saved.
2. Using the pandas library, extract the data from the file. One can use `pd.read_excel` if it is an excel sheet. Assign a variable to this action for easier use in the rest of the python script.
3. From the data, assign appropriate columns to appropriate values. For example, if your x-axes will be the years then assign the years column to some variable called `x`.

4. using either matplotlib or seaborn, assign the correct values to the correct sections in their lineplot functions (for seaborn it is `sns.lineplot()`, and for matplotlib it is `plt.plot()`).
5. Write `plt.show()` to print the graph.

4 Results

4.1 Graphical comparison of CR proton intensity and magnetic field cycles

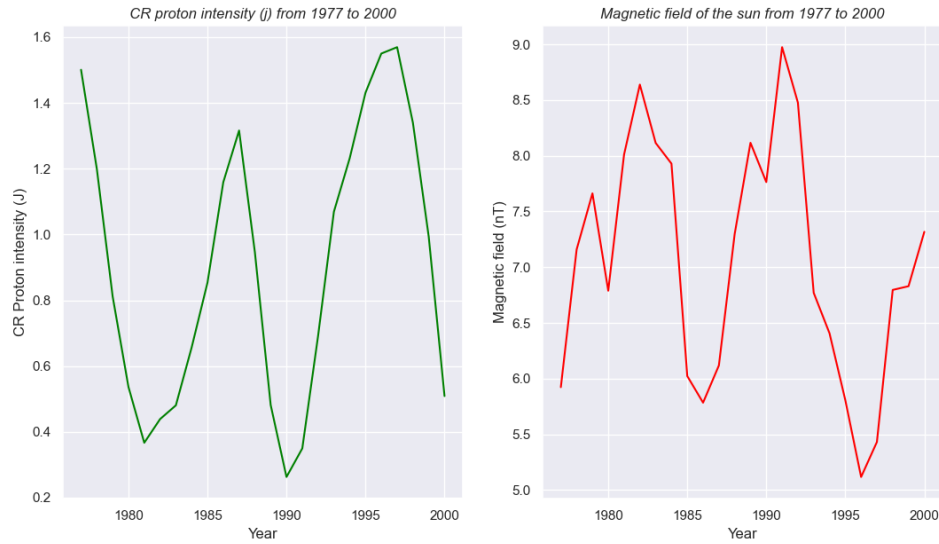


Figure 1: Two graphs illustrating the the CR proton intensity and magnetic field of the sun over the years 1977 to 2000.

4.2 Average magnetic field calculation $\overline{B}$

There are 24 values of magnetic field given within the dataset. To calculate the average, we make use of:

$$\overline{B} = \frac{\sum_{i=1}^{24} B_i}{24}$$

It can thus be shown that:

$$\overline{B} = 7.05 \text{ nT}$$

4.3 Calculating the modulation parameter (M) and plotting the parameter

It is known that the modulation parameter (M) can be expressed as:

$$M = -\ln\left(\frac{j}{j_b}\right)$$

To calculate the j_b value, use the fit of *Voyager* observations, with an r_b value of 85 AU. To be exact, the formula:

$$j_b(85AU) = \frac{17.0(P)^{-2.4}}{2.2 + 2.1(P)^{-3}} \quad (8)$$

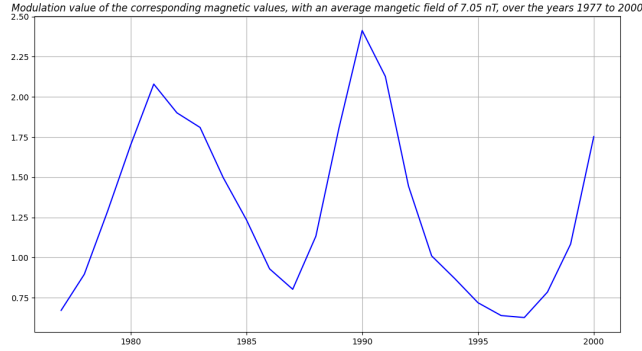
From the research of Moloto & Engelbrecht (2020), we have a P value of $P = 1.28GV$. Thus, we can conclude that:

$$j_b(85AU) = \frac{17.0(1.28)^{-2.4}}{2.2 + 2.1(1.28)^{-3}} = 2.936369787j$$

With this calculated value, it can now be plugged into the Modulation parameter formula, becoming:

$$M = -\ln\left(\frac{j}{2.936369787}\right)$$

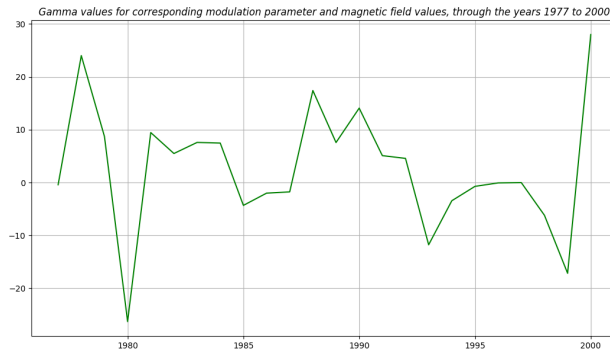
Using this formula, alongside the magnetic field values observed by the space craft, Modulation parameter values for each year can be calculated. 24 modulation values were calculated, giving the following table of values:



Year	Modulation Value
1977	0.671172471
1978	0.895649173
...	...

Table 1: Snippet of the modulation values per year

4.4 Computation of Gamma values



The average gamma value, using the same formula as for calculating the average magnetic field, yields a gamma value of:

$$\bar{\gamma} = 2.71582$$

5 Discussion

When looking at the two graphs comparing the CR proton intensity and the magnetic field of the sun from the years 1977 to 2000, there seems to be clear (almost) inverse relationship between the two values. between the years 1980 and 1985, the proton intensity was very low and the magnetic field was at a high value. Here the one phenomena can explain the other. Higher magnetic field strength can cause more solar wind being ejected from the sun, leading to more modulation upon the protons. The magnetic field could have acted as a sort of "dampener" to the protons that entered the solar system.

Now, if we examine the modulation across the years, the modulation parameter started very high in 1980, and then proceeded to lower across the 5 years. Looking at proton intensity in that same time period, it starts low but then increases, and the magnetic field starts high and drops off. This can be explained by the weakening magnetic field, which means that less energy "resists" against the CR protons upon their entry. In other words, the solar wind the sun emits is weaker than before, so the CR proton intensity will be higher.

For the gamma values, these have an almost inverse relationship with modulation values, M . For example, when we look at the year 1980 specifically, that is when the gamma value is at its lowest. But in that same year, modulation value is relatively quite high. But then, when looking at the year 1990, when gamma is quite high, modulation is also at its highest peak, alongside magnetic field. So gamma appears to have a different relationship tethered to these two phenomena than a simple direct or inverse relationship. However, the gamma value is theoretically expected to be 2, and the average is 2.7 so it seems relatively close to theoretical models.

6 Conclusion

The aim of this experiment was to more closely examine the relationship between the solar cycle, solar wind and the entering cosmic rays. And as it is clear to see, when the solar cycle is at a high activity period, magnetic fields are stronger and CR proton intensity is at a low value, and then when CR proton intensity is high, magnetic field strength is low.

And if modulation parameter values are accounted for, then it fits well into the model. Modulation refers to the changes a cosmic ray experiences due to magnetic field presence. Modulation values are low when the magnetic field values are low, meaning CR protons will experience very little changes upon their entry into the solar system.

This information is valuable to spacecraft engineers and members of aviation fields, especially those that fly crafts at particularly high altitudes. With the data here, one can more accurately predict when cosmic rays will have the most intensity and be the most dangerous to members of the aviation field given the time of the solar cycle.

7 References

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